

FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY GRADE 10 – SUB-STRAND 3.1 ACIDS AND BASES: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Remember the eight cups at the front of the class in Lesson 1: lemon juice, black tea (chai), cow's milk, Stoney Tangawizi, tap water, baking soda solution, liquid handwash soap and dilute Jik bleach. Five drops of universal indicator turned them into a rainbow. Our Driving Question was: "What invisible properties of everyday Kenyan liquids cause them to turn such different colours, and why does it matter for our health, our farms, and our environment?" In the space under each prompt, write a full scientific explanation in your own words. Use the Arrhenius definitions, ions, the pH scale, indicator behaviour and the chemical properties of acids and bases that you learned in Lessons 1 to 10. Use evidence from our experiments and give chemical equations where asked. Write in full sentences.

Part 1: Sorting the cups into groups

Look back at the eight cups from Lesson 1. (a) Sort all eight cups into three groups (acidic, neutral, basic) and state the universal indicator colour you observed for each group. Make sure every cup is placed in a group. (b) Describe how red litmus, blue litmus, phenolphthalein and methyl orange would each behave in an acid and in a base. (c) Explain why we discuss taste and feel as properties of acids and bases but never use them to test substances in the laboratory.

Write your answer here:

Part 2: What is inside an acid or a base?

Explain at the particle level why the acidic cups and the basic cups gave different colours. In your answer: (a) state the Arrhenius definition of an acid and of a base; (b) write ionisation equations for HCl, H₂SO₄, NaOH and Ca(OH)₂; (c) explain how the particles you named link to the colours of lemon juice and liquid soap with universal indicator, and to what happens in tap water.

Write your answer here:

Part 3: Why the shades differ – pH, strength and ionisation

The cups did not all show the same shade within the red group or within the blue-purple group. (a) Explain what the pH scale measures and how pH numbers relate to the indicator colours and to H^+ concentration. (b) In Lesson 9 hydrochloric acid gave pH 1 while ethanoic acid in vinegar gave about pH 3, and sodium hydroxide gave pH 13 while ammonia solution gave about pH 11. Suggest what property of the particles in these solutions could account for the differences, and show how the equations for a strong and a weak acid differ. (c) Use your ideas to explain why lemon juice (red) and Stoney Tangawizi (orange-red) differ in shade, and why the basic cups differ in shade from each other.

Write your answer here:

Part 4: What acids do – chemical properties

Acids are not only detected by colour; they also react. (a) Name the four types of substance that acids react with in the lessons, and for each give the products and the test or observation that provides the evidence. (b) Write one balanced equation for acid + metal, one for acid + carbonate, and the net ionic equation for neutralisation. (c) Explain what these reactions have in common in terms of the Arrhenius model.

Write your answer here:

Part 5: Why it matters – health, farms and environment

Answer the second half of the Driving Question. Using examples from Kenya, explain why the acid or base properties of liquids matter for (a) health, (b) farming and (c) the environment or industry. In each case name the substance, say whether it is an acid or a base, and explain the chemistry involved. Finally, write a short concluding paragraph that answers the whole Driving Question, referring to all eight cups.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Classification and use of evidence	Correctly sorts all eight cups and accurately uses indicator colours and observations from the experiments to support every claim.	Sorts most cups correctly and uses some indicator evidence, with minor errors or gaps.	Sorts few cups correctly or gives colours without linking them to acid, base or neutral.
Particle-level explanation (Arrhenius model)	States both Arrhenius definitions correctly, writes correct ionisation equations, and links H^+ and OH^- to indicator colours.	States the definitions and writes most equations correctly, but the link to the colours is incomplete.	Gives incomplete or incorrect definitions or equations, with little reference to ions.
pH, strength and ionisation	Explains the pH scale and the tenfold change, and clearly distinguishes strong from weak using degree of ionisation and correct arrows.	Explains pH and the strong/weak difference with some imprecision.	Confuses strength with concentration or pH values, or omits ionisation.
Chemical properties and equations	Gives all four reaction types with correct products, tests and balanced and ionic equations, and identifies H^+ as the common active particle.	Gives most reaction types, with minor errors in equations or tests.	Gives few reactions or has several incorrect products or equations.
Kenyan applications and final conclusion	Gives accurate Kenyan examples for health, farming and environment with correct chemistry, and a clear conclusion answering the Driving Question.	Gives relevant examples with partial chemistry and a conclusion that answers most of the Driving Question.	Gives vague or incorrect examples and a conclusion that does not address the Driving Question.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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